



2023 CSO discharges in the Wye Catchment

A report prepared on behalf of Wye Salmon Association (WSA)

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Introduction

Over the last few years, the UK water industry has been rolling out telemetric sensor technology to monitor the discharge status¹ of CSOs on the sewer network. This is often referred to as EDM (event duration monitoring). In 2021, we published a report (Green, 2021) detailing results from 2019-20 pertaining to the River Wye Catchment. We now have EDM data available for 2021, 2022 & 2023 which allows us to examine the trends. Is anything getting better?

I have downloaded datasets for England & Wales in 2020, 2021, 2022 & 2023 as GIS files from the [Rivers Trust](#) website and spatially filtered these to examine emissions falling within the Wye Management Catchment². This combined data forms the basis of the present work.

A rising trend apparent in 2019-20 was in part driven by an increase in the number of sites monitored. However, by 2020, substantially all DCWW CSOs had been fitted with EDM. Reporting rates are not 100% however, as there are always a few EDMs not functioning for one reason or another. But reporting is generally around 95% - high enough not to impact the big picture.

I begin by looking at the national trends. The raw numbers from the last couple of years have been much reported in the media. It has been noted that the number of discharges fell slightly in the 2022 data -which was dismissed by some as being the result of the dry summer. Whether this assertion is based on science or arm-waving is not clear. The substantial increase in national emissions observed in 2023 has, in turn, been blamed by the water industry on high rainfall. I shall examine these assertions for the Wye MC data.

National picture

During 2019, EDM deployment was still ongoing, but by 2020 coverage had reached 95% of all CSOs. However, by 2022, the fraction has fallen to 92%. Although the number of EDM deployments has continued to increase, the number of *known* CSOs has increased faster. It seems the need to publish EDM data has shone a light into the murky world of undocumented CSOs. Moreover, it has exposed the fact that significant number of CSOs are unpermitted; meaning not registered with EA/NRW and thus operating entirely illegally. Neither the water industry, nor the environment agencies appear to have been terribly concerned by this omission prior to the onset of the EDM program.

Across England and Wales, in 2023, there were **579,581** reported CSO discharges, with a total duration of **4,608,495 hours**, from a total of **16,558** sites. In terms of duration, this equates to **2x the 2022 value**.

Because the different water companies differ in size, it is useful to consider this in terms of the average values of *number of discharges per site* and *total duration per site* as KPIs. Figures 1 & 2 below show some selected trends over the 4-year period.

The black bar indicates the industry average across England and Wales. Both charts show a progressive improvement over 2020-22, with a huge deterioration in 2023.

¹ EDM only reports the duration of discharge; sadly, it does not record volume of discharge. Clearly an hour of discharge from Llangurig pumping station is rather different to an hour from the Eign CSO in Hereford.

² The Wye MC is the full Wye catchment down to the estuary, as defined by EA & NRW.

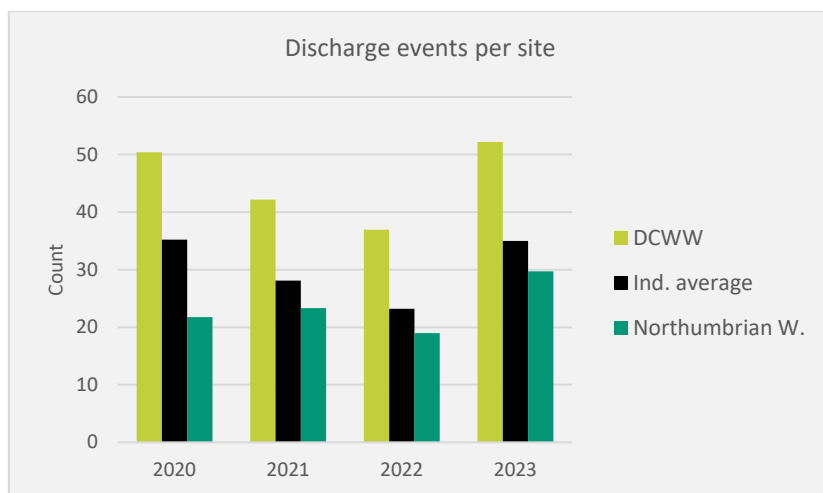


Figure 1: average number of discharges per site for England & Wales

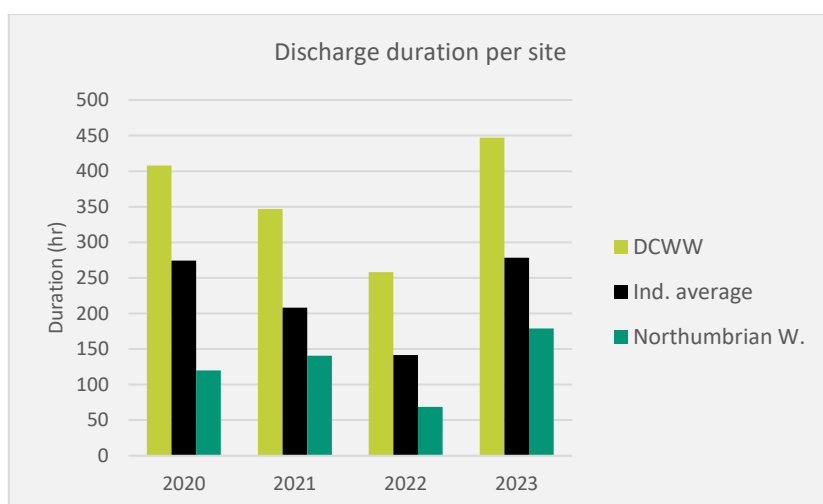


Figure 2: average total discharge duration per site for England & Wales

Figures 1 & 2 also include data for two specific water companies; DCWW is included because of its relevance to the Wye catchment. Throughout the 4-year period, DCWW has been a consistently poor performer overall, falling well above the industry average. Out of the 10 water companies, it had the highest values for both KPIs in 2021, 2022 & 2023. In 2020, it was marginally outdone by Thames Water and United Utilities. The charts also include data for Northumbrian Water, a company which consistently outperform the industry average. Northumbrian Water outperform DCWW by roughly a factor of two. The reasons for this disparity are not clear, but both investment history and geography are likely to play a part. Across Wales and the Borders, **DCWW** was responsible for **117,830 spills** in 2023, with a total duration of **1,009,450 hours**: about a quarter of the England & Wales total.

Wye Management Catchment

From the above national data, we now apply a spatial filter to home in on the Wye MC. Figure 3 below shows the locations of known CSOs within this area. As of 2023, these numbered 121, of which 8 remained unpermitted; meaning that they have no legal right to emit anything at all into the rivers. But they do. In 2023 the Wye MC was subjected to **5,827 spills**, totalling **50,075 hours**.

The size of the circles in Figure 3 is a rough indication of the total discharge duration at each site. It is already clear that there are truly awful emission durations in the headwaters, at Llangurig, Llanwrtyd Wells and Beulah (just north of Llanwrtyd Wells). We shall return to these.

Figures 4 & 5 below show our two KPIs for the Wye MC.

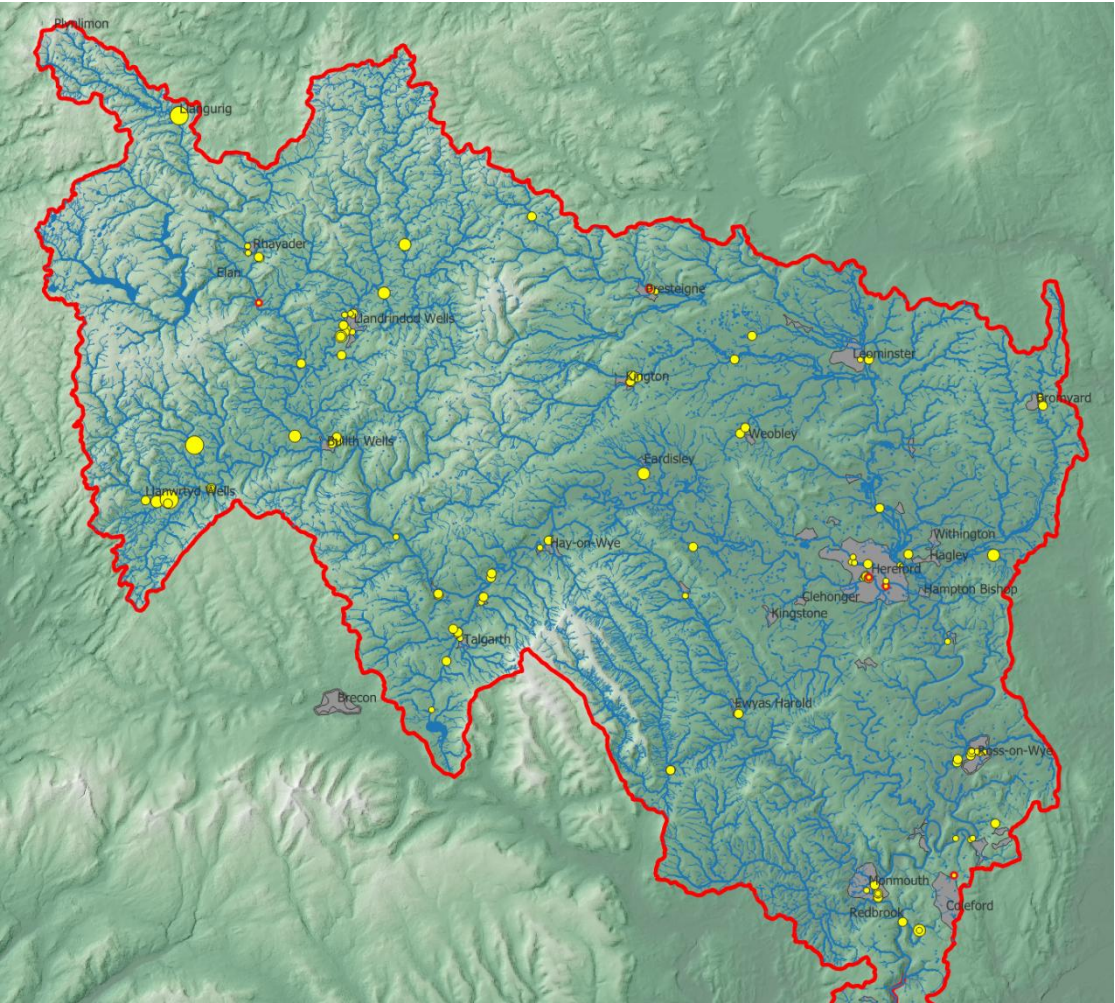


Figure 3: CSOs (yellow) located within the Wye MC (red outline). Size of circle indicates spill duration, ranging from <50 hours to >2000 hours (e.g. Llangurig). Red halos indicate sites with a dysfunctional EDM.

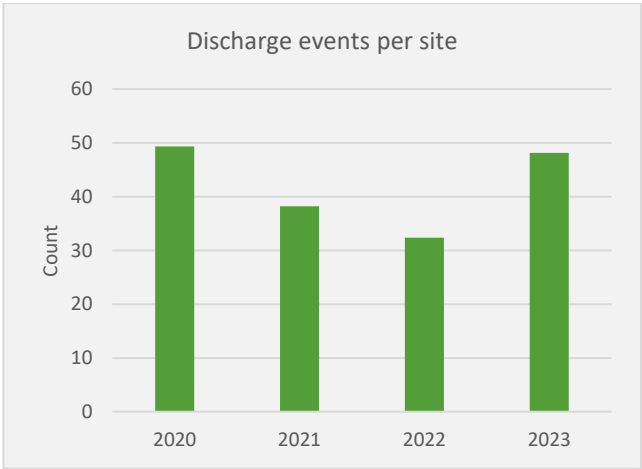


Figure 4: average number of spills per site within the Wye MC

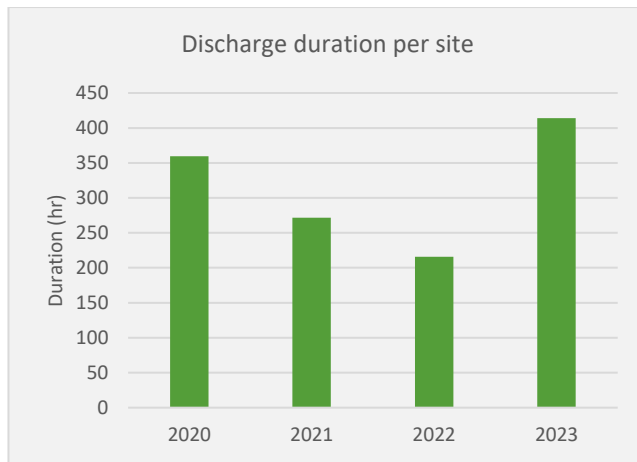


Figure 5: average discharge duration per site within the Wye MC

The first observation is that both charts are a close match to the overall DCWW performance shown in Figures 1 & 2. The Wye SAC clearly gets no special treatment regarding the emission of untreated sewage.

Secondly, the improving trend seen over the period 2020-22 was completely reversed in 2023. Let's now look at rainfall. The river catchment is an excellent integrator of rainfall, so we can use a convenient EA monitoring station on the main stem as a proxy. Figure 6 shows the winter (October-March) flow volume logged at Belmont over the same 4-year period. We find a spookily similar chart.

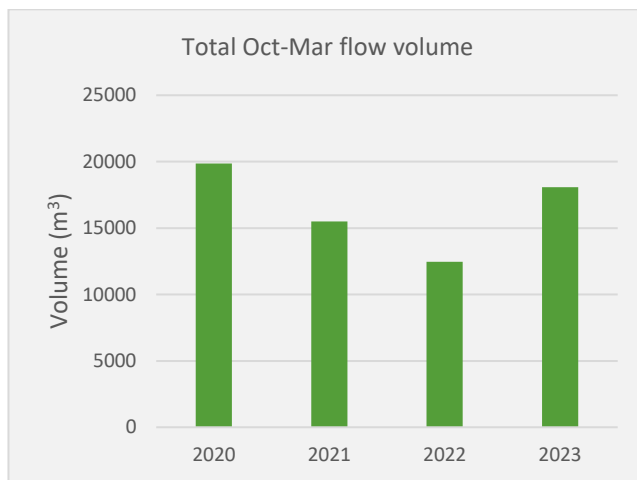


Figure 6: winter (Oct-Mar) flow volume on the Wye at Belmont, just above Hereford.

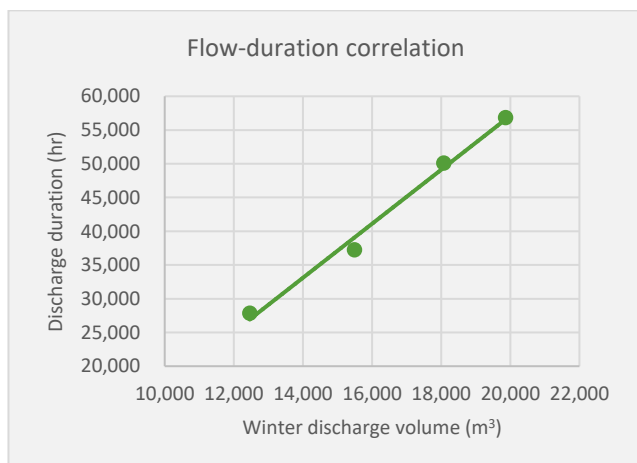


Figure 7: correlation between total CSO discharge duration within the Wye MC and the winter flow volume at Belmont (above Hereford), for the 4 year 2020-23. The regression line has $R^2 = 0.99$.

We can check this out objectively by looking at the correlation. Figure 7 shows a scatter plot of the total CSO discharge duration within the Wye MC versus the winter flow volume at Belmont. It appears to follow a simple linear relationship, with a very high R^2 value, confirming that discharges are probably responding primarily to rainfall.

Worst performing sites in 2023

The number of sites within the Wye SAC which exceeded 1,000 hours of discharge in 2023 is 14: too many to list here. In table 1 below are listed the sites which exceeded around 2,000 hours of discharge, which roughly equates to 12 weeks. One would think that these, the most egregious sites, would be targeted by DCWW for urgent improvement. But apparently not.

Site name	2020 spill duration	2023 spill duration
Pwllmeyric SPS (CSO)	3,556	5,756
Llanwrtyd Wells CSO on Trunk Sewer	2,592	3,290
Beulah SPS Beulah Powys	2,957	3,125
Llangurig P.S. - Overflow	191	2,739
Abernant Lake CSO	2,064	1,967

Table 1: top 5 sites for discharge duration in 2023

Pwllmeyric SPS discharges into the Mouton Brook (off the bottom of Figure 1). Although falling within the Wye MC, it is not part of the Wye catchment, discharging directly into the Severn estuary just west of Chepstow. As such, it is arguably of less concern to us. Nevertheless, it has been malfunctioning for the last 4 years, apparently without intervention, and is indicative of the wider situation.

Llanwrtyd Wells discharges into the upper reaches of the Irfon, part of the Wye SAC, and prime salmon spawning territory. Again, we find 4 years of ongoing malfunction. Furthermore, Llanwrtyd Wells, with a population of around 1,875 is a significant settlement. A trunk sewer overflow here potentially represents a large pollution load going into the upper Irfon.

Beulah lies a few miles north-east of Llanwrtyd Wells and discharges into the Cammarch, a tributary of the Irfon which is also part of the Wye SAC and salmon spawning territory. Beulah is a smaller settlement, of maybe 200 people, but the Cammarch is also small, and this duration of sewage discharge is likely to represent a serious pollution load to the river. Once again, we find no improvement over 4 years.

Llangurig PS discharges into the upper reaches of the Wye. As above, although Llangurig is a small settlement, the Wye is also small up here and continuous discharges or raw sewage of this duration are probably not insignificant to the river ecology. The performance of this site has degraded progressively over the 4-year period, increasing from 191 hours in 2020, to 372 hours in 2021, 852 hours in 2022, and 2,739 hours in 2023.

Abernant Lake CSO is also part of the Llanwrtyd Wells sewer system, discharging directly into the upper Irfon. And, somewhat predictably now, we find no improvement over 4 years.

Conclusion

- On the basis of both number and duration of discharges per listed site in 2023, DCWW remains the worst performing water company across England and Wales.
- Discharges within the Wye MC are typical of the wider DCWW picture. The SAC does not get special treatment regarding discharges of untreated sewage.
- Reducing emissions during 2021-22, and the significant increase in 2023 are, indeed, entirely explainable by year-to-year variations in winter rainfall.
- Overall, the data for the Wye MC shows no indication of any attempt by DCWW to reduce untreated sewage emissions over the 4-year period from 2020-23. The linear relationship between winter rainfall and sewer discharge duration is suggestive of widespread under-capacity of the sewage system.
- Examination of the five worst performing sites within the Wye MC also suggests that there has been no meaningful attempt to fix the problems. This is especially worrying, given the ecological sensitivity of the SAC headwater streams receiving the discharges from four of these sites.